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Crashworthiness study of a monopile OWT with metamaterial-based fenders under vessel collisions

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ABSTRACT

The growing risk of vessel collisions with offshore wind turbines (OWTs) poses significant threats to both maritime safety and wind farm economics. To mitigate the consequences upon accidental collisions, this study proposes a metamaterial-based fender concept that effectively protects both OWT structures and impacting vessels. Key variables such as impact velocities and operational wind loads are considered. Special emphasis is placed on the unique characteristics of OWT-vessel collisions, distinguishing them from traditional waterway structure impacts. A case study is carried out for a 22-MW OWT and a 5000 deadweight tonnage vessel with impact velocities ranging from 2 to 6 m/s. Nonlinear finite element simulations are performed for the impact scenarios. Through the simulations, we demonstrate the effectiveness of the metamaterial-based fender design. The structural damage of the OWT is reduced to negligible levels and nacelle accelerations remain within the safety threshold of 6 m/s^2. For the striking vessel, the proposed fender design decreases the crushing depths of the bow by 14.53%, thereby improving post-impact survivability. Additionally, the influence of the wind speed and its relative angle to the impact direction of the vessel proves to be a critical factor in collision responses. Moreover, varying metamaterial geometric configurations are examined, revealing a nonlinear relationship between cell density, structural stiffness, and overall OWT protection performance. Future research will focus on optimizing fender designs for specific scenarios to balance risk, efficiency, and economic feasibility.

1. Introduction

Offshore wind turbines (OWTs) are being installed at an increasing rate worldwide [1,2]. However, their presence can pose obstacles to maritime navigation and increase the risk of vessel collisions. For example, a bulk carrier heavily collided with a monopile OWT in the Hollandse Kust Zuid wind farm in January 2022. In September 2024, the service operation vessel collided with a monopile OWT at the Ørsted Hornsea wind farm in the North Sea, damaging the base of the OWT. Zhang et al. [3] reviewed vessel-OWT collision accidents and highlighted the rising frequency, which is projected to occur 1.5 to 2.5 times annually [4]. Such high-energy impacts can cause severe structural damage and disrupt turbine operations due to induced vibrations and accelerations.

To evaluate the dynamic responses of OWTs under impact, Song et al. [5] conducted detailed research on collisions between a ship and a 5-megawatt (MW) monopile OWT under various wind conditions. The results indicate that the turbine may collapse into the sea or fall onto the ship's deck, leading to a secondary impact. Yu et al. [6] studied ship collisions with a 10-MW floating offshore wind turbine (FOWT) and found that, although the FOWT can absorb more impact energy than fixed OWTs due to its floating characteristics, the nacelle may suffer damage from excessive acceleration, and large pitch motions following the impact could potentially lead to capsizing. Yu et al. [7] also reviewed structural responses and design considerations for offshore tubular structures subjected to ship impacts.

From an economic perspective, any accidental structural damages are unacceptable. Moreover, the possible consequences for collided vessels, such as oil spills or loss of human life, are intolerable. Offshore repair operations are also significantly more complex and costly compared to those for inland structures. Therefore, it is imperative to implement robust protective measures to mitigate such collision damage and safeguard both the OWTs and ships.

Zhang et al. [3] reviewed existing protective technologies for waterway structures against vessel collisions, listing five main passive anti-collision measures which have been widely applied to bridges. Among these options, floating thin-wall structure fenders (Table 5

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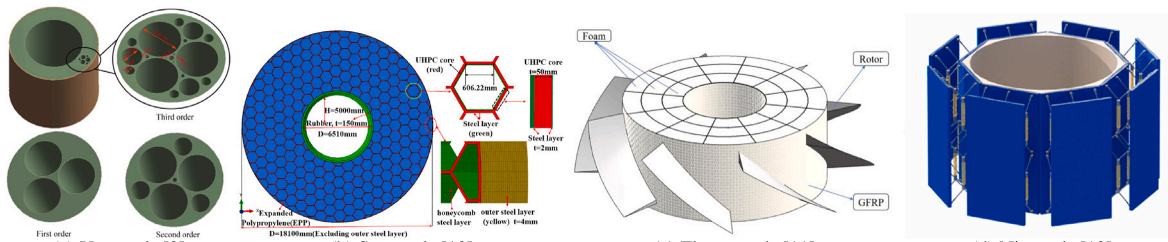


(a) Yue et al. [9] (b) Sun et al. [10] (c) Zhang et al. [11] (d) Nie et al. [13]

Fig. 1. The previous different anti-collision fender designs for OWT.

in [3]) are identified as the most suitable and feasible choice for collision protection of OWTs, considering the offshore environment, OWT structure shapes, water level changes, and navigation requirements.

Given the outstanding profile of floating thin-wall fenders, subsequent research has concentrated on refining their geometry, testing alternative material and structural combinations. Han et al. [8] designed protective fenders with various geometrical dimensions and materials to safeguard a fixed OWT with a tripod foundation from collisions with a 2500 t ship. The influence and cost-effectiveness of different geometry and material combinations were analyzed. Based on that, Yue et al. [9] proposed protective fenders for fractal pores, which found that the first-order fractal significantly improves the resistance to impact, see Fig. 1(a). Sun et al. [10] incorporated composite honeycomb structures as internal components within the fenders, as shown in Fig. 1(b). A finite element analysis (FEA) and an elasto-plastic analysis were conducted to study the energy dissipation mechanism. Zhang et al. [11] developed a rotational fender designed to redirect incoming ships as indicated in Fig. 1(c), reducing the energy absorption required by the fender. Compared with other designs, this design minimizes the damage of OWTs, ships, and fender, resulting in potentially higher reusability. Although this concept is initially proposed for bridges [3,12], this design may be particularly well-suited for OWTs due to their circular shape. Nie et al. [13] proposed a spring system as the internal structure illustrated in Fig. 1(d), enabling high stiffness under low general loads and low stiffness during high-impact collisions. This design enhanced the structural stiffness and damping, leading to reduced response peaks and suppression of transient vibrations.

Nowadays, novel metamaterials have attracted growing attention for their effects in mitigation of impact consequences. Metamaterials are a class of artificially engineered structures that exhibit extraordinary electromagnetic, acoustic, or mechanical properties beyond those of their base materials. These properties arise from carefully designed structures rather than the intrinsic characteristics of their constituent materials. Most metamaterial-based structures enhance crashworthiness by increasing specific energy absorption [14–16], promoting elastic deformation [17], or leveraging wave attenuation mechanisms through low-frequency bandgaps [18]. Another group provides cloaking functionalities by manipulating wave propagation, widely applied in optical, acoustic, thermal, electrical, and elastic domains [19]. An impact event induces stress waves in solids, which also propagate as elastic waves. However, compared to acoustic waves, elastic wave cloaking is less developed due to the complexity [20].

A major limitation of current protective fenders is that energy absorption mainly relies on the impact region, leading to stress concentration, premature failure, and inefficient utilization of the fenders' full capacity [3]. Integrating wave cloaking concepts into anti-impact design could enhance both safety and durability by redistributing stress more effectively. Rather than achieving complete elastic wave cloaking, the objective is to improve overall fender performance by partially guiding stress waves toward non-impact regions and reducing transmission to protected structures. This concept is particularly suitable for monopile OWT, which has a circular structure and houses vulnerable electrical and mechanical components inside the nacelle. These components are highly sensitive to excessive accelerations [21–24].

Therefore, developing protective fenders that integrate metamaterial-based stress-diversion mechanisms is crucial. Such designs not only safeguard structural integrity but also mitigate extreme accelerations, minimizing economic loss. To meet these requirements, the present study designs an annular metamaterial-based OWT fender with stress diversion capabilities and evaluates its crashworthiness under various vessel impact conditions. This research emphasizes not only the structural integrity but also its operational continuity, ensuring that high-energy collisions do not compromise its functionality. Through comprehensive numerical simulations and parametric analyses, the effectiveness of the proposed fender is evaluated, offering valuable insights for the advanced protective systems. The structure of this paper is organized as follows: Section 1 outlines the research motivation and background; Section 2 proposes the metamaterial-based fender concept; Section 3 details the OWT and impact vessel selection, and collision scenario design for case study; Section 4 provides the numerical FEA modeling setups; Section 5 conducts detailed post-impact analysis from various perspectives; Section 6 summarizes the general conclusions.

2. The proposed metamaterial-based fender concept

The design of an engineering application fender must account for a wide range of considerations, including protective performance, manufacturing and installation feasibility, self-floating capability, and economic competitiveness. This section presents two innovative fender configurations for OWT protection: (i) a baseline metamaterial fender and (ii) a ring-enhanced variant.

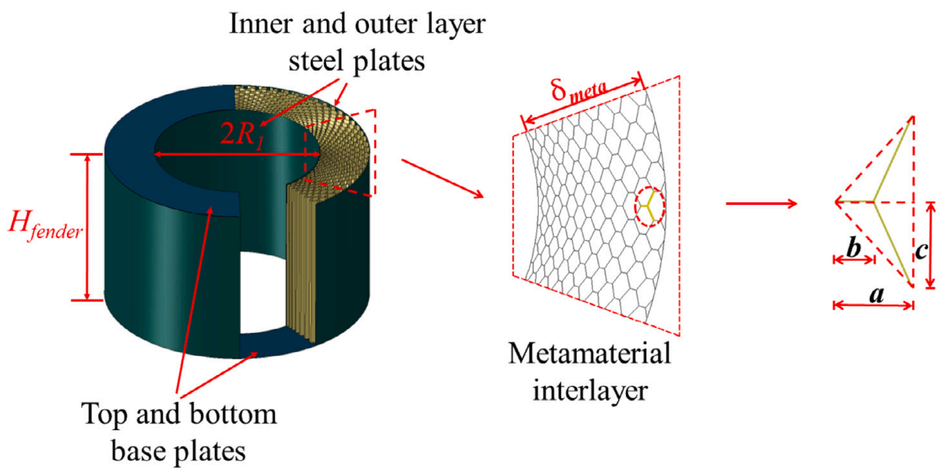
2.1. Design of baseline metamaterial fender

To accommodate the variation in sea levels and corresponding changes in collision heights, a self-floating annular metamaterial-based protective fender—hereinafter referred to as "metamaterial fender"—is proposed, with its configuration illustrated in Fig. 2(a).

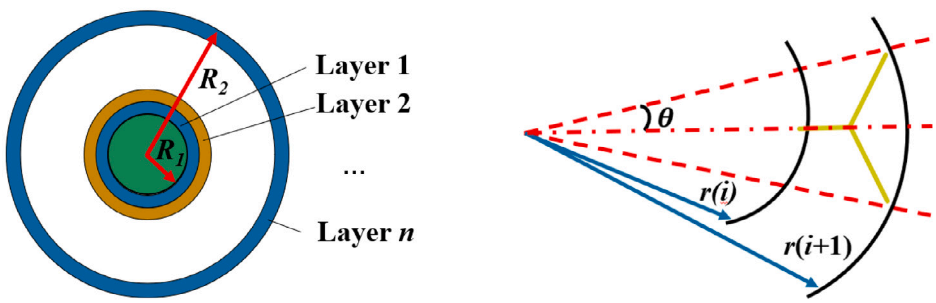
2.1.1. Global configuration

The fender is designed with a total height of H_{fender}, an inner diameter of $2R_1$, an outer diameter of $2R_2$, and a metamaterial core part with a thickness of δ_{meta}, as illustrated in Fig. 2(a). The height is determined by balancing buoyant force, structural self-weight, and the vertical extent of potential vessel impact zones. The inner diameter is set according to the outer diameter of the OWT tower to ensure proper installation. Structurally, the fender comprises an inner steel shell, an outer steel shell, top and bottom cover plates, and an intermediate metamaterial core.

The outer, top, and bottom cover plates ensure the watertightness of the structure and make it easier to manufacture and assemble. For this type of annular structure, it is typically divided into 2–4 segments during practical production and installation. The presence of these outer plates facilitates transportation, welding, and other processes.



(a) Global fender structure and the internal metamaterial units



(b) Layer design parameters for the metamaterial

Fig. 2. Global fender and internal structures of the proposed metamaterial fender concept.

2.1.2. Internal metamaterial structure

The internal metamaterial structure of this presented fender is derived from an acoustic cloaking design based on the pentamode metamaterial (PM) principle [25]. The concept of pentamode materials was first proposed by Milton and Cherkaev in 1995 [26]. The work described a class of metamaterials where only a single non-zero eigenvalue exists among the six eigenvalues of their elastic stiffness matrix. This unique characteristic imparts a fluid-like property, rendering them highly effective for designing cellular structures capable of guiding stress waves along predetermined paths. Previous scaled experiments have confirmed that this form exhibits effective stress wave cloaking performance under point impacts [27]. Considering the stress wave cloaking performance and manufacturing feasibility, this metamaterial configuration, illustrated in Fig. 2(a), has been adopted as the core of the presented anti-collision fender. The acoustic wave equation for a PM structure with anisotropic modulus [28] is expressed as:

$$K\mathbf{S} : \nabla\left(\rho^{-1}\mathbf{S}\nabla p\right) - \ddot{p} = 0\,, \tag{1}$$

where K and ρ denote the bulk modulus and density of the PM material, respectively; S is the elasticity tensor matrix, $\mathrm{div}(S) = 0$, and p represents the sound pressure. The modulus matrix C of PM [28] can be expressed as:

$$\mathbf{C} = K\mathbf{S} \otimes \mathbf{S}\,. \tag{2}$$

Following the basic principle of acoustic cloaking, a linear radial mapping function is defined as:

$$f(r) = \frac{R_2 - \eta R_1}{R_2 - R_1} r + \frac{R_1 R_2}{R_2 - R_1}(\eta - 1)\,, \tag{3}$$

R_1 and R_2 are inner and outer radius of ring-shaped acoustic cloaking, r is the starting radius of each layer of acoustic cloaking, η is acoustic cloaking effect ($0 < \eta < 1$). Assuming the bulk modulus and density of the external medium are $k_w = 1$ and $\rho_w = 1$, respectively [28], and the ratio of the radius of the outer fender to the radius of the inner fender is defined as k_R ($R_2 = k_R \cdot R_1$), the physical parameters for 2D PM cloaking can be calculated as:

$$K = \frac{\left(k_R - 1\right)^2 r}{\left(k_R - \eta\right)^2 r + k_R(\eta - 1)\left(k_R - \eta\right) R_1}\,. \tag{4}$$

As the designed 2D PM cloaking is ring-shaped, for a certain point $P(r, 0)$, we have:

$$\mathbf{S} = \begin{bmatrix} \dfrac{\left(k_R - \eta\right) r + k_R(\eta - 1)R_1}{\left(k_R - 1\right) r} & 0 \\ 0 & \dfrac{k_R - \eta}{k_R - 1} \end{bmatrix}, \tag{5}$$

then by introducing Eqs. (4) and (5) into Eq. (2):

$$\hat{C} = \begin{bmatrix} 1 + \dfrac{k_R(\eta - 1)R_1}{\left(k_R - \eta\right) r} & 1 & 0 \\ 1 & \left[1 + \dfrac{k_R(\eta - 1)R_1}{\left(k_R - \eta\right) r}\right]^{-1} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{6}$$

In the 2D case, PM cloaking can be reduced to bimode metamaterial (BM) cloaking. The modulus matrix of a BM microstructure takes the

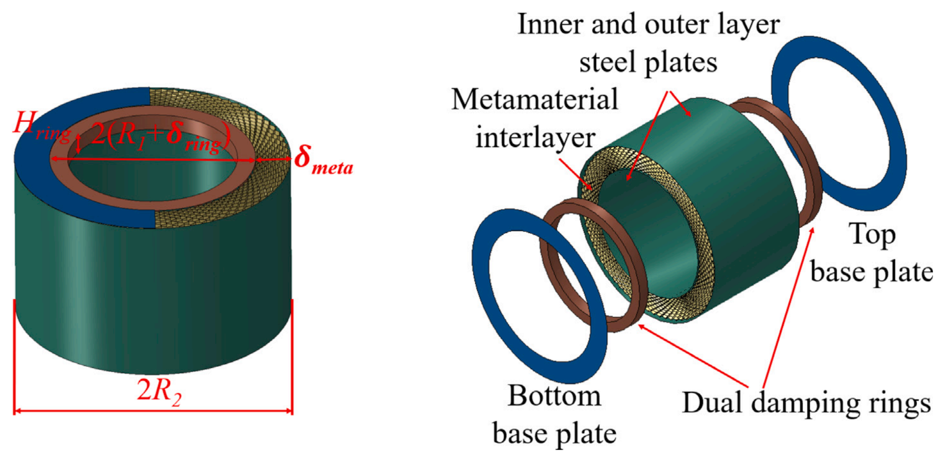


Fig. 3. The diagram of dual damping ring-enhanced meta fender.

form [26]:

$$\mathbf{C}' = K' \begin{pmatrix} a-b & 0 \\ 0 & 1/a \end{pmatrix} \otimes \begin{pmatrix} a-b & 0 \\ 0 & 1/a \end{pmatrix}, \tag{7}$$

which leads to:

$$\hat{C}' = \frac{a-b}{a} K' \begin{bmatrix} a(a-b) & 1 & 0 \\ 1 & [a(a-b)]^{-1} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{8}$$

By comparing Eqs. (6) and (8), we derive the following geometric relationships for the unit cells of the presented metamaterial [26]:

$$\begin{cases} \dfrac{a(i)}{c(i)} = \sqrt{2 - \dfrac{2k_R(1-\eta)R_1}{(k_R-\eta)\,r(i)}} \\ b(i) = \dfrac{a(i)}{2} \end{cases}, i = 1, 2, 3 \ldots n. \tag{9}$$

The layered metamaterial structure of the fender is determined by three key geometric parameters. The circumferential division angle θ. For the present design η is 0.3 and the circumferential angle is $\theta = \frac{\pi}{72}$, which divides the fender into 72 units circumferentially, see Fig. 2(b). The metamaterial is composed of 3-link unit cells, as illustrated in Fig. 2(a). The geometry of each cell is defined by parameters a, b, and c ($0 < b < 2a$). While the size of the unit cells varies between layers, their shape ratio is maintained, governed by the relationship in Eq. (9).

The geometric parameters for each layer are determined by a progressive calculation. To ensure geometric continuity between adjacent layers, an equal sector angle design is implemented. The geometric parameters for each layer are calculated recursively using the following relations:

$$\begin{cases} r(i+1) = r(i) + a(i) \\ c(i) = r(i+1)\sin\theta \end{cases}, i = 1, 2, 3 \ldots n. \tag{10}$$

Compared with more complex metamaterials such as chiral or hierarchical heterogeneous structures, which often require advanced techniques like laser melting or additive manufacturing, the proposed design, made of simply stacked steel plates, can be produced using conventional methods such as butt welding, offering significantly greater manufacturing feasibility.

2.2. Design of the ring enhancement

Due to the multiple-layer metamaterial design, these layers are expected to suffer from crushing during collision and lead to some superimposed structural vibrations. Such impact excitation, occurring relatively close to the fixed end (tower base), will inevitably propagate toward the free end (tower top), ultimately resulting in a motion response at the tower top, manifested as high acceleration. This forecast is supported by the observed high-order fluctuations in the internal energy of the OWT and the acceleration waveform patterns under the original metamaterial protection in the later analysis.

The core problem lies in the ineffective attenuation of vibration transmission from the impacted fender to the OWT structure. In non-integrated connected systems, vibrations and accelerations are transmitted through physical connections; the more direct and rigid these connections are, the more efficiently vibration energy is transferred [29,30]. Therefore, reducing the area of direct physical connections between structures and incorporating lower-stiffness components at interface links [31,32] can effectively achieve stiffness decoupling. This strategy isolates the protected structure more from the vibration source, thereby mitigating transmitted accelerations.

In response to this vibration control principle, the original design has been modified as illustrated in Fig. 3. Dual damping rings integrated at the OWT-fender interface and a reduced-thickness metamaterial layer maintain the original metamaterial fender thickness. Similar damping design concepts have been widely employed before in such anti-impact fenders for structures [3,10,33–37]. In the present research, both rubber and aluminum foam (Al foam) are employed as dual damping ring materials to enhance the vibration suppression performance of the proposed metamaterial fender concept. The two material components have been widely applied to modify anti-impact structures' damping systems. These configurations are referred to as "dual rubber damping ring-enhanced meta fender" and the "dual Al foam damping ring-enhanced meta fender", hereafter abbreviated as "rubber ring-enhanced fender" and "Al foam ring-enhanced fender", respectively. The ring geometry is controlled by its radial depth δ_{ring} and its height H_{ring} as depicted in Fig. 3.

3. Case study

To evaluate the effectiveness of the proposed metamaterial-based fender concept, a set of case studies is conducted. A representative OWT, the IEA 22 MW OWT, is selected as the target prototype due to its large scale and relevance to future offshore energy developments. A 5000 DWT vessel is chosen as the striking body, representing a typical medium-sized ship frequently involved in maritime operations near OWTs. Based on these prototypes, the geometric and structural parameters of the fender are defined accordingly. Additionally, key collision variables including impact velocities and wind loads are outlined at the end of this section to support subsequent numerical investigations.

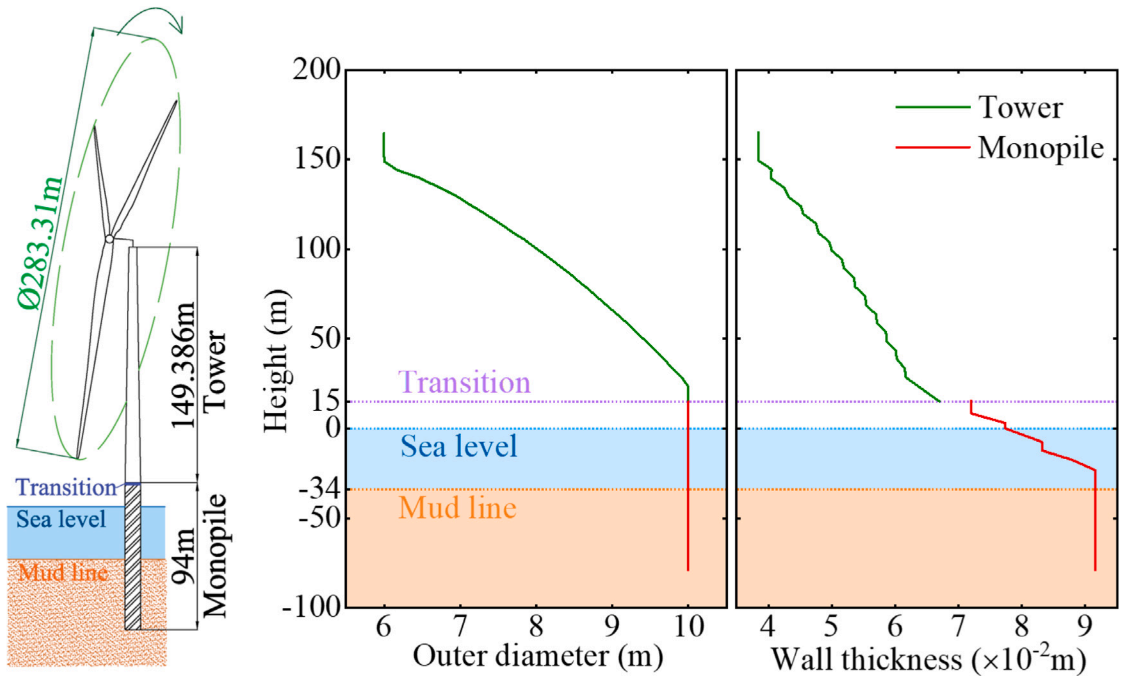


Fig. 4. Sketch of the IEA 22 MW OWT, and the distribution of outer diameter and wall thickness for the tower and monopile.
Source: Redrawn from [39].

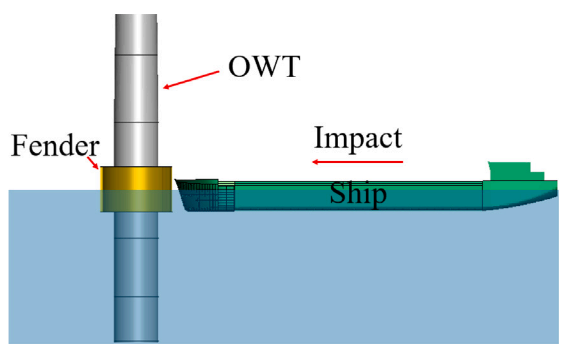


Fig. 5. Schematic of vessel-OWT-fender collision system (side view).

3.1. IEA 22MW OWT prototype

The offshore wind energy market has seen continuous upscaling in turbine power ratings and rotor sizes over the past decade, encompassing both fixed and floating types. In this study, considering the complex coupled motions of floating OWTs, the fact that most current collision accidents have occurred with fixed OWTs, and the early-stage nature of fender design, the IEA 22-MW turbine with a monopile foundation is selected. This turbine, the largest offshore wind turbine prototype as of 2025 [38], was jointly developed by DTU and NREL [39].

The IEA 22-MW OWT features a rotor-nacelle assembly with a total mass of 1215.60 t, a tower mass of 1574.00 t, and a monopile mass of 2097.00 t. The tower extends 149.39 m in length and tarts from 15.00 m above mean sea level. The monopile, with a total length of 94.00 m, reaches the tower base and has 45.00 m embedded in the seabed as shown in Fig. 4. The assembly also includes a transition piece between the monopile and tower, which is modeled as a simple point mass of 100.00 t. The distributions of outer diameter and wall thickness along the height are presented in Fig. 4.

3.2. Striking vessel

In this study, a bulk carrier with a 5000 DWT—previously adopted in related research [40,41]—is selected as the striking vessel. This vessel class is defined by the NORSOK N-004 standard [42] as representative of typical OWT-ship collision scenarios. The vessel's principal dimensions include an overall length of 86.00 m, A moulded breadth of 15.00 m, A moulded depth of 6.50 m, and A design draft of 5.00 m. To account for hydrodynamic effects during a head-on collision, an added mass coefficient of 0.25 is applied, according to DNV recommendations [43].

3.3. Parameters of the proposed fenders

As previously introduced in Section 2, and based on the geometric specifications of the IEA 22 MW OWT and the striking vessel, the fender parameters are explicitly defined in this study. All variants are constructed with standard Q235 steel, reducing construction expenses and enhancing cost competitiveness, and feature a metamaterial layer sheet thickness is $T_1 = 0.004$ m, inner and outer cover plate thickness is $T_2 = 0.004$ m and top/bottom plate thickness is $T_3 = 0.002$ m.

For the original metamaterial fender design, the total height H_{fender} is set to 10.00 m. The inner diameter $2R_1$ is 10.20 m—slightly larger than the 10.00 m outer diameter of the OWT tower to facilitate practical installation. The outer diameter $2R_2$ is 16.20 m, resulting in a metamaterial thickness δ_{meta} of 3.00 m.

Constructed with standard Q235 steel, the proposed metamaterial fender configuration attains a total mass of 267.12 t and a volume of 1244.07 m^3, resulting in an overall structural density of 214.72 $\mathrm{kg/m^3}$. This design achieves a draft of 2.09 m through buoyant equilibrium, which can be further adjusted by applying additional ballast loads.

Regarding the ring-enhanced fender variants, the outer diameter $2R_2$ is maintained at 16.20 m to allow for consistent comparative analysis. In these designs, the damping rings with a thickness δ_{ring} of 1.00 m and a height H_{ring} of 1.00 m are integrated, as illustrated in Fig. 3, yielding a total mass of 210.12 t and a volume of 892.39 m^3, structural density of 220.96 $\mathrm{kg/m^3}$, and draft of 2.05 m achieved through self-buoyancy.

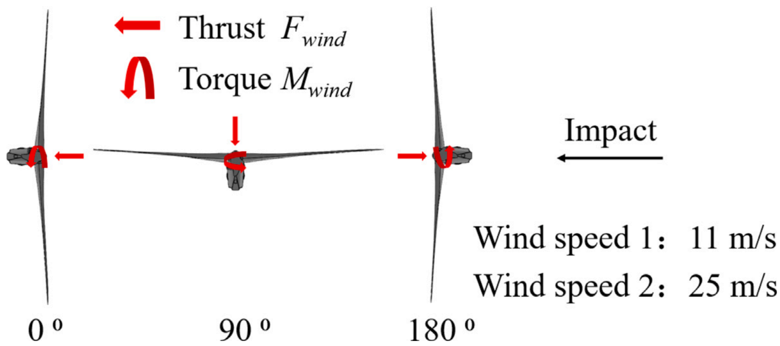


Fig. 6. Schematic of the relative angle between vessel impact and wind loads (top view).

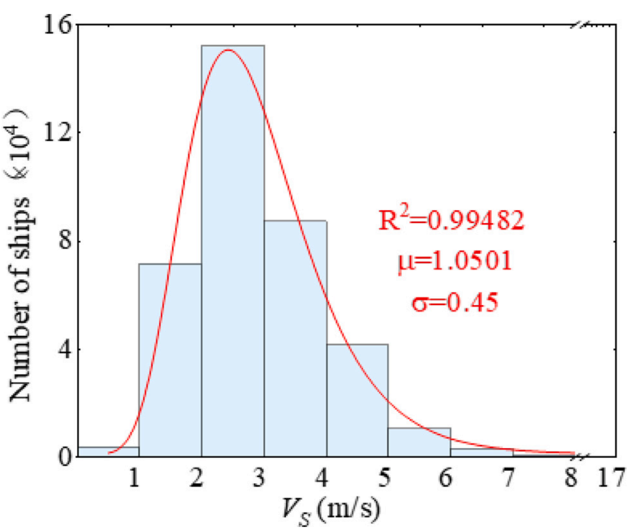


Fig. 7. Probability distribution of the ship passing bridge velocity [41].

Table 1
Variables in collision scenarios.

Vessel mass (t)	Added mass (t)	Impact velocities (m/s)	Wind load speeds (m/s)	Relative angle between wind and impact (°)
6800	1700	2, 4, 6	0, 11, 25	0, 90, 180

3.4. Collision scenario

Existing international regulations, such as Rule 6 of the International Regulations for Preventing Collisions at Sea [44], do not stipulate explicit speed limits for vessels navigating near offshore structures. Instead, they require that “every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions”.

The special statistical research on the maritime ship passing velocities is also limited. In previous research on the vessel-OWT collision, most of them conducted collision analysis under relatively low vessel velocity ranges, e.g. 1–3 m/s [5], 0.5–3 m/s [45], 2, 3 m/s [6], 2 m/s [9], 0.5–2.5 m/s [11], 1.5, 2 m/s [13], 1–4 m/s [10], 1–5 m/s [46].

However, some insights can be drawn from inland waterway transportation data. For example, Zhang et al. [40,41] compiled velocity statistics for ships passing under a bridge on the Yangtze River, indicating that the majority of vessels operate at speeds below 6 m/s as shown in Fig. 7. Hence, this study primarily investigates high-energy collision scenarios, with a maximum impact velocity of 6 m/s and a corresponding collision energy of 153 megajoules (MJ). A head-on impact configuration is analyzed to represent the most severe loading condition.

Four distinct collision scenarios under different protections are systematically compared: (a) OWT with no protection; (b) OWT protected by the proposed original metamaterial fender; (c) OWT protected by the rubber ring-enhanced fender; (d) OWT protected by the Al foam ring-enhanced fender.

Furthermore, the above scenarios are conducted in the OWT parked conditions without wind loads; however, the collision almost always happens coupled with wind loads [5]. Hence, two critical wind regimes are evaluated: 11 m/s (rated wind speed) and 25 m/s (cut-out wind speed), as defined in the IEA 22 MW reference turbine specifications [39]. The aerodynamic forces and moments acting on the OWT are derived from computational results of the aeroelastic model developed by Collier et al. [47]. Three characteristic wind-impact angle configurations are investigated, as shown in Fig. 6: (i) 0° alignment (co-directional): Wind and impact forces act in the same direction; (ii) 90° perpendicular: Wind loads apply lateral bending moments during transverse impact; (iii) 180° opposition (counter-directional): Wind and impact forces create competing load vectors. All the variables in the present collision scenarios are summarized in Table 1.

4. Finite element modeling and numerical setup

The Finite element (FE) modeling details and numerical setup are described in this section, especially the OWT and the impact vessel FE model and their contact definition.

4.1. Vessel FE model

The vessel’s bow is modeled in detail using shell elements with a refined mesh size of 0.1 m to capture the impact dynamics accurately. To improve computational efficiency, the rest of the vessel’s structure, which is far from the impact area, is modeled with rigid shell elements and a coarser mesh size of 0.6 m. The complete vessel model comprises a total of 199 212 elements. The FE model of the vessel’s bow is presented in Fig. 8.

The elastic–plastic deformation behavior of the vessel’s Q235 steel bow structure, incorporating strain rate effects, is governed by the Cowper–Symonds constitutive model (Eq. (11)). This model is employed to accurately simulate the dynamic crushing response of the bow during impact, with the specific material parameters detailed in Table 2. Since the rest of the vessel is almost not involved during collisions, the hull and stern are modeled as a rigid body to improve computational efficiency.

$$\frac{\sigma_d}{\sigma_s} = 1 + \left(\frac{\dot{\varepsilon}}{C}\right)^{\frac{1}{P}} \tag{11}$$

where σ_d is the dynamic yield strength, σ_s is the static yield strength, $\dot{\varepsilon}$ is the strain rate, C and P are the strain rate parameters.

Table 2
Mechanical properties of main materials employed in the presented research.

Rubber and Al foam				Steel			
Parameter	Unit	Rubber	Al foam	Parameter	Unit	Q235	S345
Density	kg/m^3	1080	178	Density	kg/m^3	7850	7850
Young's modulus	MPa	/	370	Young's modulus	GPa	210	200
Poisson's ratio	/	0.4995	0.3	Poisson's ratio	/	0.3	0.3
C_{10}	MPa	0.7925	/	Yield strength	MPa	285	345
C_{01}	MPa	0.432	/	Tensile strength	MPa	/	450
σ_{pl}	MPa	/	3	Shear stiffness	GPa	/	79.3
γ	MPa	/	2	Compressive strength	MPa	/	450
α_2	MPa	/	93.57	C	/	40.4	/
α	/	/	0.2	P	/	5	/
β	/	/	5.79	Fatigue slope	/	/	3
ε_D	/	/	6	Failure strain	/	/	0.3

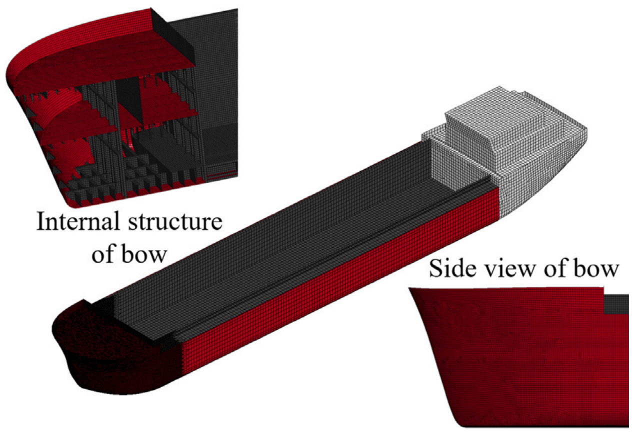


Fig. 8. Impact vessel FE model.

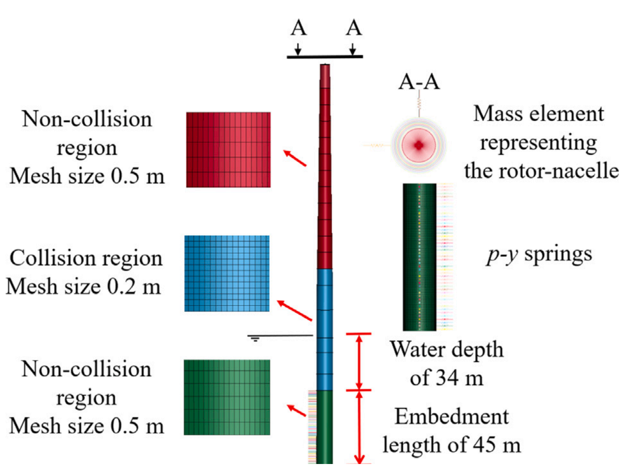


Fig. 9. OWT FE model.

4.2. OWT FE model

Details of the 22 MW OWT are presented in Section 3.1. The FE model in the present study employs Belytschko–Tsay shell elements with five through-thickness integration points to account for strain rate effects. A refined mesh of 0.2 m is applied in the collision regions, while a coarser 0.5 m mesh is used elsewhere to save computational time, as illustrated in Fig. 9. The numerical model comprises 113 738 shell elements, with the nacelle and rotor blades simplified as a lumped mass at the tower top to reduce computational complexity. To account for the flexibility of the monopile foundation, the soil–structure interaction is modeled using distributed springs with depth-dependent stiffness properties [23]. The spring constants are derived through linearization of the standard $p-y$ curves. Each spring element is connected between two nodes: one node is rigidly linked to the monopile structure, while the opposite node is fully fixed in all translational and rotational degrees of freedom. The monopile foundation is constrained at its base against vertical translation (z-direction) and rotation about the z-axis. The tower and monopile are modeled using an elasto-plastic material model for steel S345, the material's stress–strain constitutive relation is given as follows:

$$\sigma = k\left(\varepsilon_{yp} + \bar{\varepsilon}^{p}\right)^{n} \tag{12}$$

$$\varepsilon_{yp} = \left(\frac{\sigma_y}{k}\right)^{\frac{1}{n}} \tag{13}$$

where σ_y is the yield stress, ε_{py} is the elastic strain to yield, $\bar{\varepsilon}^{p}$ is the effective plastic strain, k is the strength coefficient and n is the hardening exponent. The corresponding mechanical properties are listed in Table 2.

The natural frequency of a monopile-based OWT plays a critical role in determining its dynamic response during collision events. In this study, which focuses on evaluating the global structural response and deformation of OWTs under vessel impact, the first-order natural frequency of the simplified OWT model is validated against the reference value provided in the definition document [39]. The first fore-aft or side-natural frequency of the simplified model is 0.1673 Hz, while the reference value is 0.1600 Hz. The simplified model demonstrates excellent agreement (4.6% relative difference) with the reference while maintaining consistent fixed-base boundary conditions identical to the detailed reference model.

4.3. Numerical modeling of the fender systems

The fender geometries are described in Section 2. Shell elements (0.2 m element size) are employed for the steel components, and solid elements (0.2 m element size) are utilized for both rubber and Al foam constituents. The total element counts are determined to be 233 410, 227 336, and 227 336 for the metamaterial fender, rubber ring-enhanced fender, and Al foam ring-enhanced fender, respectively. The rubber material behavior is simulated using the Mooney–Rivlin constitutive model:

$$W = C_{10}\left(I_1 - 3\right) + C_{01}\left(I_2 - 3\right) + C\left(I_3^{-2} - 1\right) + D\left(I_3 - 1\right)^2, \tag{14}$$

$$C = \frac{\left(C_{10} + 2C_{01}\right)}{2}, \tag{15}$$

$$D = \frac{C_{10}(5\upsilon - 2) + C_{01}(11\upsilon - 5)}{2(1 - 2\upsilon)}, \tag{16}$$

where I_1, I_2, I_3 are the first, second, and third deviatoric strain invariants of right Cauchy–Green tensor. The coefficients C and D are related to C_{10} and C_{01} as shown in Eqs. (15) and (16), with D being a function of the material's Poisson's ratio, υ. C_{10} and C_{01} are

Mooney–Rivlin coefficients, typically calibrated from the stress–strain curves or correlated to the hardness of rubber. In this study, the specific values for these coefficients are adopted directly from the work of Liu et al. [21].

The mechanical behavior of Al foam is characterized by the foam material constitutive model proposed by Deshpande and Fleck [48]:

$$\sigma_Y = \sigma_{pl} + \gamma \frac{\hat{\varepsilon}}{\varepsilon_D} + \alpha_2 \ln\left(\frac{1}{1-\left(\frac{\hat{\varepsilon}}{\varepsilon_D}\right)^{\beta}}\right), \tag{17}$$

where σ_{pl} is flow stress, ε_D is the locking strain, $\hat{\varepsilon}$ is the effective strain and γ is the linear hardening modulus, α is the yield surface shape parameter which controls the yield surface shape, α_2 is the nonlinear hardening modulus, and β is the nonlinear hardening parameter. The specific values for these parameters are listed in Table 2.

4.4. Global numerical settings

In this study, a separation distance of 0.6 m is established between the vessel's bow and the protective structure (or unprotected OWT) to prevent initial contact, with the vessel assigned an initial velocity of 6 m/s as depicted in Fig. 5. The interaction between the vessel bow and the protective structure is defined using the contact-automatic-surface-to-surface command. Internal structural contacts for both the vessel bow and the protective structure are specified with the contact-automatic-single-surface command. A friction coefficient of 0.2 is assigned for the interface between the vessel bow and the protective structure, as well as for the internal structural contacts [8].

4.5. Performance evaluation criteria for fender systems

The protective performance of the fender systems is evaluated through five quantitative metrics, addressing both structural integrity and operational safety:

1. OWT crushing depth (τ): Maximum permanent deformation of the OWT, measured as the displacement of the most deformed node from its pre-impact position.
2. OWT damaged area (A_W): Area along the OWT (vertical range: −11 m to +15 m) where plastic strain exceeds 0.001, indicating irreversible material failure.
3. Bow crushing depth (λ): Maximum permanent indentation depth of the vessel's bow structure post-impact.
4. Watertight bulkhead maximum deformation (ζ): Maximum permanent deformation of the vessel's watertight bulkhead.
5. Watertight bulkhead damaged area (A_S): Area of the watertight bulkhead where plastic strain exceeds 0.001, compromising compartment integrity.

Additionally, nacelle acceleration is constrained to a maximum threshold of 6 m/s^2 under extreme collision conditions, aligning with IEC guidelines for parked or emergency states [21,49,50]. This criterion ensures protection of precision nacelle components (e.g., drivetrain, electronics) from excessive lateral accelerations—a critical concern distinct from conventional waterfront structures.

4.6. Validation of the finite element metamaterial model

To validate the metamaterial performance, a numerical simulation replicating the hammer impact test reported in [27] was conducted. The test targeted the same metamaterial structure design, and its setup is illustrated in Fig. 10(a). The specimen was fabricated from Q235 steel, consistent with the present fender design, with its mechanical properties summarized in Table 2. The experimental configuration was reproduced in the numerical model, including two fixed constraint points, a vertical hammer impact, and five monitoring points located within the inner layer. The measured hammer force signal was extracted and applied as a time-varying concentrated load in the simulation, enabling direct quantitative comparison. The structure consists of 13 layers comprising 50 pentamode lattice unit cells with defined geometry (R_1 = 52 mm, R_2 = 104 mm, t = 1 mm, h = 10 mm). The mesh was discretized with quadrilateral shell elements of 1 mm, yielding 89 480 elements in total. Strain monitoring points (A–E) were positioned along the inner layer to match the experimental sensor locations shown in Fig. 10, ensuring synchronized data collection for validation.

Table 3
Peak strain measurement validation between experiments and simulations.

Measurement points	Peak strain ($\mu(\times10^{-6})$)					
	A	B	C	D	E	A/E
Experiment [27]	101.3	−56.8	−123.2	29.6	97.5	1.04
Present simulation	91.1	−52.4	−114.7	31.3	91.9	0.99
Relative errors (%)	−10.1	−7.8	−6.9	5.8	−5.7	4.8

As shown in Table 3, strain levels at point A (front surface) and point E (rear surface) are comparable and both lower than the peak strain at point C. This distribution indicates effective redistribution of stress waves away from the direct impact zone. The numerical results agree well with the experiments, with peak strain deviations within 10%, confirming the reliability of the methodology for subsequent analyses.

5. Results and analysis

The analysis is divided into four subparts within this section: (i) A maximum-speed head-on vessel impact scenario without wind loads, focusing on the protective performance comparison among the proposed fenders to identify the optimal design; (ii) A post-impact response assessment under lower impact velocities, aimed at evaluating fender performance in non-extreme conditions and exploring their potential for reuse in general applications; (iii) Collision scenarios coupled with wind loads, varying with wind speeds and the relative angles, to investigate the combined post-impact response. (iv) Collision scenarios incorporating varying metamaterial geometric configurations, to quantify their influences on structural integrity.

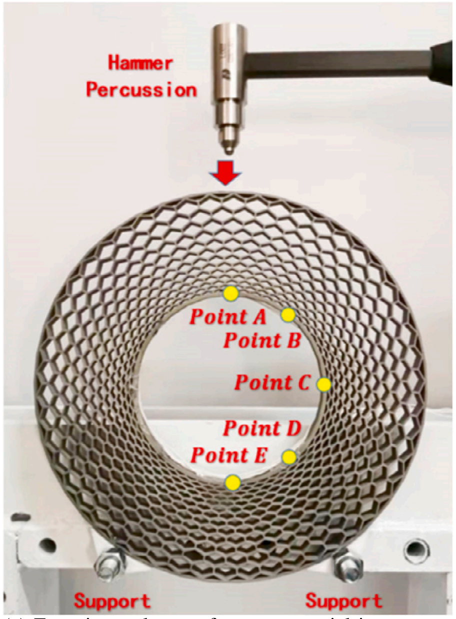
5.1. Post-impact response under different proposed fenders

This subsection presents a comprehensive analysis of the head-on vessel impact at 6 m/s without wind loads, examining impact force, energy transformation, acceleration, stress distribution, and structural deformation to compare the performance of different fender designs and identify the optimal configuration.

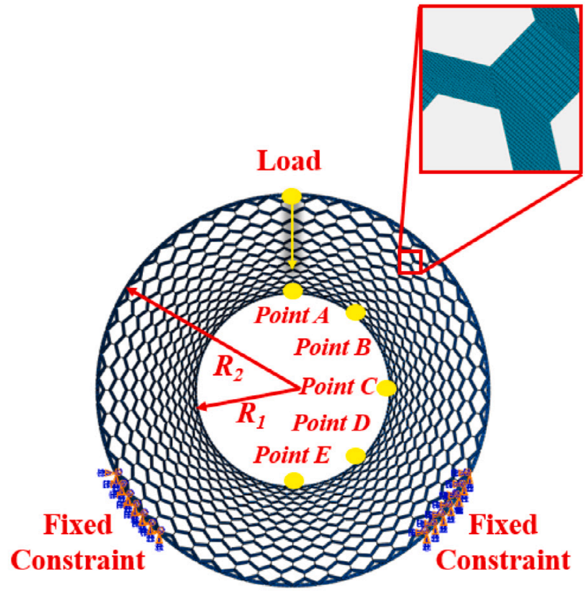
5.1.1. Impact force

The impact force histories on OWT across the proposed fenders and with no protection condition are compared in Fig. 11. Force fluctuations are consistently observed across all cases, resulting from structural failure mechanisms during the impact events, mainly generated by the failure of the vessel's bow.

Both rubber and Al foam ring-enhanced fenders demonstrate certain peak force reduction capabilities, decreasing the maximum values from the original metamaterial fender's 50.4 MN to 44.2 MN (12.3% reduction) and 44.8 MN (11.1% reduction), respectively. Notably, these enhanced fender systems outperform the unprotected case (49.1 MN), achieving 9.8-10.0% maximum force reduction relative to no protection.



(a) Experimental setup for metamaterial impact test [27].



(b) Numerical setup for the metamaterial validation

Fig. 10. Experimental and numerical setup for metamaterial validation.

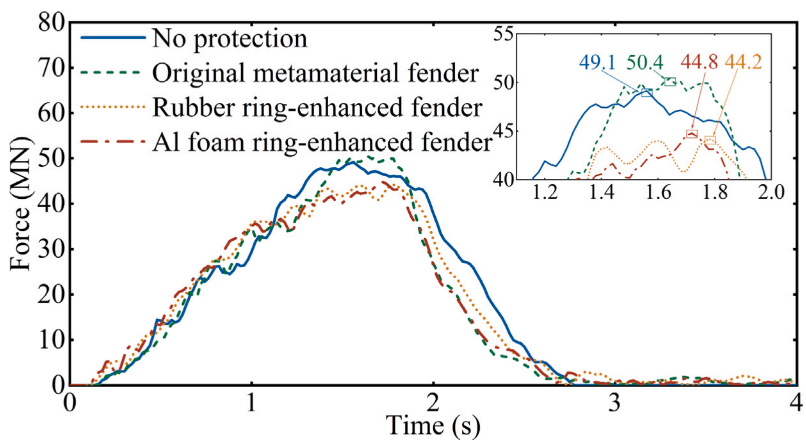


Fig. 11. The impact force duration of different protections after optimization.

5.1.2. Energy transformation

The whole energy transformations during and after collision with various collision systems are presented in Fig. 12. All the curves demonstrate excellent energy balance throughout the simulation duration, verifying numerical stability and conservation principles. During the collision, the initial kinetic energy is transformed into the internal energy, sliding energy, hourglass energy, damping energy and remaining kinetic energy. The limited hourglass energy contribution (consistently <1.25% of internal energy across all cases) and tight energy balance confirm the simulation's numerical reliability for collision analysis.

The internal energy transformation histories among various components in each collision system are compared in Fig. 13. Despite the reduced radial width of the metamaterial core part in the ring-enhanced fenders, their general energy absorption capacity has slightly increased compared to the original metamaterial fender design, reaching 35.5% and 33.6% of the total energy.

All fender configurations exhibit effective protective functions for both the OWT and the impacting vessel bow, as indicated by internal energy absorption and corroborated by subsequent damage analysis. For the OWT, a rapid initial energy accumulation followed by an oscillatory decay is observed—attributable to elastic energy storage and release mechanisms. In the unprotected scenario, the OWT absorbs a maximum of 46.1 MJ, approximately four times higher than in the cases with metamaterial-based fenders.

Regarding the vessel's bow, the rubber ring-enhanced fender provides the best protection, reducing internal energy absorption by approximately 40 MJ. The original metamaterial fender and the Al foam ring-enhanced fender offer slightly lower protective performance in comparison.

Concurrently, although the inner damping rings solely absorb a marginal portion of the total impact energy, they effectively contribute to vibration suppression within the OWT structure. This can be inferred from the reduced high-frequency fluctuations observed in the OWT internal energy curves. Under the protection of the ring-enhanced fenders, the internal energy curves of the OWT retain a similar first-order modal response, while higher-frequency vibrations are significantly damped, indicating improved acceleration suppression, which can be reflected from Fig. 14.

A comprehensive analysis of the internal energy histories across all protection configurations reveals that fender design plays a critical

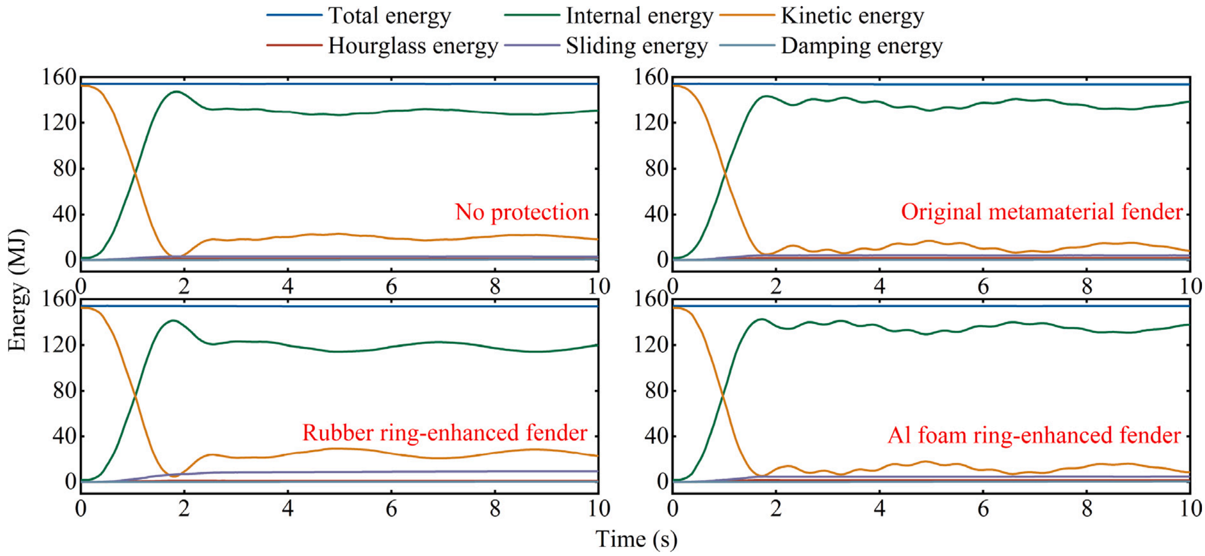


Fig. 12. The whole energy time histories of various collision systems.

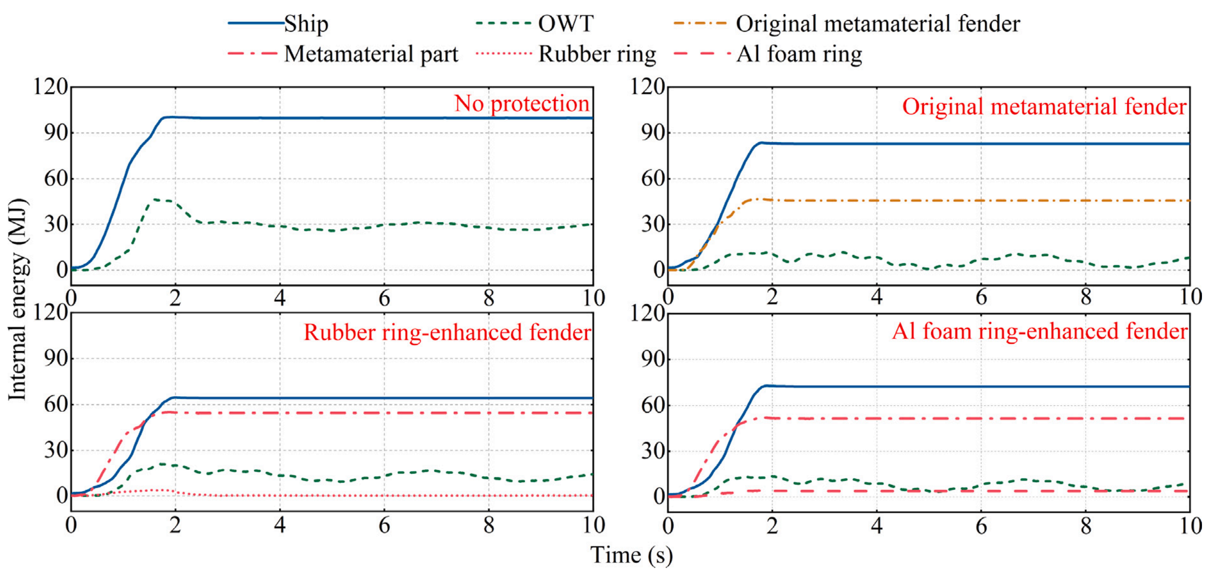


Fig. 13. The internal energy transformation of different components in each collision system.

role in energy distribution. The proposed metamaterial-based fender concept demonstrates an effective balance, mitigating damage to both OWT and the impacting vessel. This aligns with the core principle of protective design: sacrificing the protective device itself to preserve both the protected structure and the impacting object, thereby minimizing financial loss and risk to human life [3]. These differences in internal energy distribution are reflected in the deformation characteristics and response mechanisms of the structures involved, which will be further analyzed in the following sections.

5.1.3. Nacelle acceleration

Unlike other waterway structures such as bridges, OWTs contain a large number of precision-integrated components within the nacelle located at the top of the tower. Sudden lateral impacts, such as vessel collisions, can induce excessive accelerations in the nacelle, potentially leading to significant damage. Although a universally standard extreme acceleration threshold for OWT nacelles has not yet been established, operational guidelines typically restrict tower-top accelerations to a range of $0.2\sim0.4g$ during normal operation [49,51,52]. But under the parked or extreme conditions, e.g., earthquake and collisions, a higher limit is often considered acceptable, typically around $0.6g$ [49,50] or 6 $\mathrm{m/s^2}$ [21]. Therefore, in this study, an acceleration threshold of 6 $\mathrm{m/s^2}$ is adopted as the extreme criterion to evaluate the protective performance of fender designs.

The acceleration histories of the OWT nacelle under various protection are compared in Fig. 14. The response histories can be divided into two distinct phases: (I) Impact phase: characterized by forced vibration while collision forces are still acting during contact; (II) Damped free vibration phase: occurring after the vessel loses contact with the OWT.

Peak accelerations mainly occur during Phase II, except for the original metamaterial fender, where the peak arises near impact termination. This phenomenon can be explained by two primary mechanisms: First, a portion of the collision energy is dissipated through plastic deformation in the bow and the impacted structures, while the remaining OWT internal energy forms as elastic potential energy is released at the OWT's natural frequency during free vibration. Second, the massive tower's inertial response is not fully excited by the transient impact duration ($<\sim3$ s), whereas complete modal development is permitted during free vibration. This dynamic amplification effect is recognized

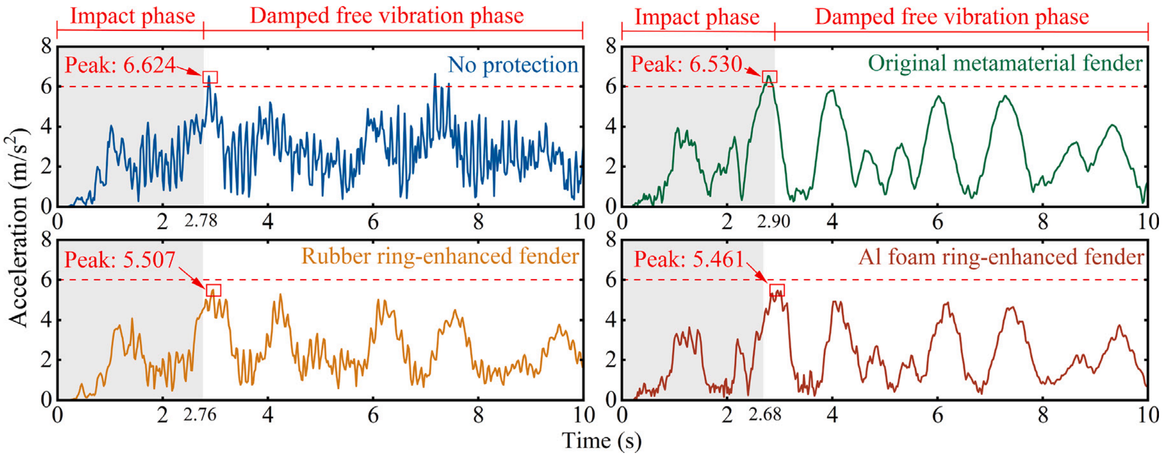


Fig. 14. The acceleration time histories of the OWT nacelle of various collision systems.

as characteristic behavior of flexible structures under transient impact conditions.

For all tested configurations, the acceleration time histories exhibit similar waveform patterns and peak structures, which correspond closely with the fluctuations observed in the OWT internal energy curves in Fig. 13. This similar waveform is due to the use of the same vessel bow, which demonstrates consistent crushing behavior [53].

The maximum nacelle accelerations recorded are 6.624, 6.530, 5.507, and 5.461 m/s^2 for the unprotected configuration, original metamaterial fender, rubber ring-enhanced fender, and Al foam ring-enhanced fender, respectively. Both ring-enhanced configurations successfully limit the peak acceleration below the preset safety threshold of 6 m/s^2. The maximum accelerations reduced 15.6% and 16.5%, respectively, compared to the original metamaterial fender (6.530 m/s^2). Which corresponds to the smoother OWT internal energy histories in Fig. 13.

Additionally, all three fender types effectively suppress the high-frequency vibration spikes observed in the unprotected scenario. This is primarily attributed to the non-integrated configuration between the fenders and the OWT structure. The presence of a non-rigid connection inhibits the direct transmission of high-frequency impact forces, thereby mitigating sharp fluctuations in acceleration.

Another important consideration is that research on OWT collisions should not be limited to the impact phase alone. It is essential to extend simulation durations to capture the subsequent damped free vibration response, during which even larger acceleration peaks may occur, for example, 10 s in the current study, 30 s by Song et al. [5], and even 100 s by Yu et al. [6].

5.1.4. *Stress and deformation*

For the OWT, the stress distributions at the time point that suffers from maximum impact forces are presented in Fig. 15 and the post-impact damages are depicted in Fig. 16 & Table 4. Despite the better performance on the acceleration suppression of the two ring-enhanced fenders, their protections to OWT structures in current configurations are weakened, the extreme stress exceeds the OWT yield strength (345 MPa) occurred in both ring-enhanced fenders around the bottom ring connections. This excessive stress causes certain plastic deformations in the OWT as shown in Fig. 16 & Table 4, although they still mitigate the damage area and depth effectively.

As illustrated in Fig. 16(b), the original metamaterial fender provides effective protection to the OWT tower, achieving a crushing depth of $\tau = 0$ m and maintaining a residual margin of allowable deformation. However, after incorporating the damping ring modifications discussed in Section 2.2, the post-impact damage shown in Fig. 16(c) and (d) reveals increased crushing depths and the onset of plastic deformation in the OWT. This degradation in protective performance is primarily attributed to a reduction in overall fender stiffness caused by the diminished radial width of the metamaterial core, resulting in greater transmission of impact energy to the OWT structure, even though it is not obvious from Fig. 13. Among the two ring-enhanced configurations, the Al foam ring-enhanced fender outperforms the rubber ring variant, exhibiting approximately half the damage depth and affected area in the OWT.

Table 4
Post-impact damage comparison of various fenders at 10 s.

Collision scenario	Maximum OWT plastic strain	OWT damage area (m^2)
No protection	0.1427	96.315
Original metamaterial fender	0	0
Rubber ring-enhanced fender	0.0665	37.306
Al foam ring-enhanced fender	0.0309	16.624

Nevertheless, the relationship between OWT damage and fender stiffness is observed to be highly nonlinear. Parametric tests of alternative ring-enhanced configurations – including increased radial depth of the metamaterial layer and higher unit densities – did not yield a monotonic improvement in protective performance. For brevity, these supplementary test results are not detailed in this work but are identified as a subject for our future investigation.

In the context of vessel-OWT collisions, it is essential to consider not only the acceleration experienced by the OWT nacelle but also the safety of the impacting vessel. In vessel–bridge collision studies, the research emphasis is typically placed on protecting bridge structures, often overlooking vessel safety. This prioritization is justified by the fact that bridges usually carry a large number of pedestrians and vehicles, and structural failure could result in mass casualties [3]. In contrast, OWT do not host human occupants or traffic; the consequences of their damage are primarily economic. However, in vessel–OWT collision scenarios, especially under real marine conditions, severe damage to vessels may directly endanger lives. Therefore, protective designs for vessel–OWT interactions must give heightened consideration to vessel safety alongside structural protection of the OWT.

To assess the post-impact vessel safety, detailed damage evaluations to the vessel bow are depicted in Figs. 17, 18, and 19, focusing on the integrity of the bulkhead in the impact bow.

The post-impact deformations and bow crushing depths are illustrated in Fig. 17. Distinct differences in the post-impact bow profiles can be observed: the unprotected and original metamaterial fender cases exhibit more circular deformation patterns, whereas the ring-enhanced fender scenarios display a flattened deformation shape. This difference corresponds to the fender deformation behavior shown in Fig. 16, where the presence of the hollow region allows more fender

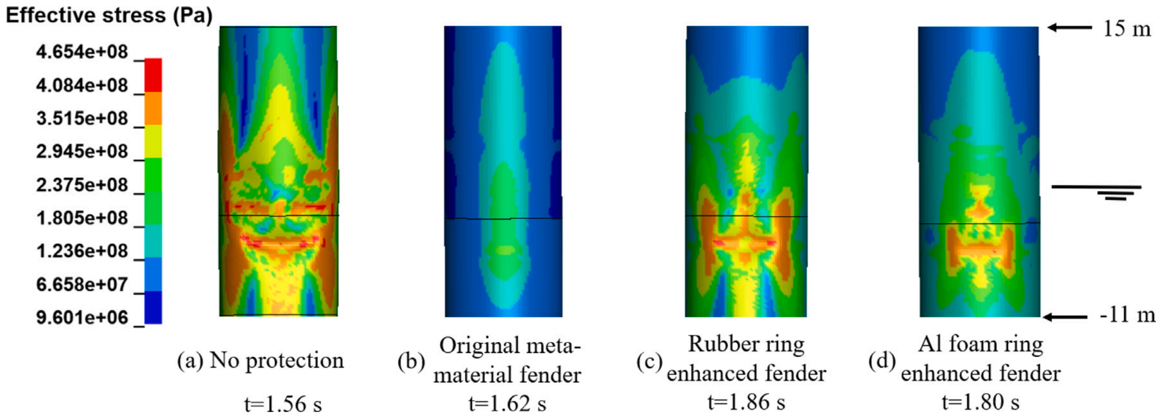


Fig. 15. OWT stress at the time point suffering from maximum impact force.

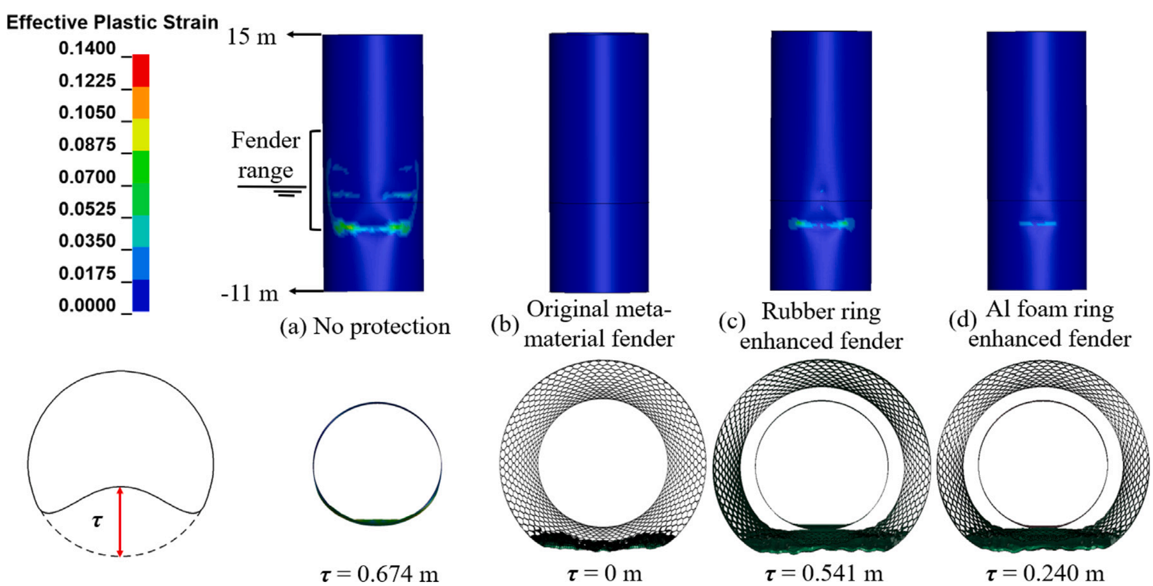


Fig. 16. Post-impact OWT damage comparison with different fenders at 10 s.

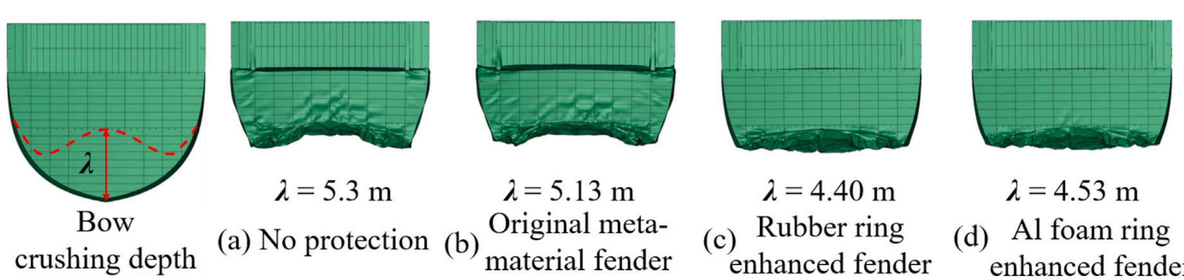


Fig. 17. Post-impact bow indentations and crushing depths in various collision systems.

components to engage in the collision, resulting in a more uniformly distributed crushing response and higher energy absorption as shown in Fig. 13. This behavior is particularly meaningful, as it significantly reduces the bow's localized crushing depth, thereby lowering the risk of damage to critical structural elements such as the bulkhead, further evidenced in Figs. 18 and 19, and ultimately reducing the probability of vessel sinking.

The post-impact deformations of the vessel bow equipped with various fenders are illustrated in Figs. 18 and 19. The investigated vessel bow section is equipped with two vertical watertight bulkheads, referred to as the 1st and 2nd bulkheads in the following analysis.

In the scenario where the vessel collides directly with the OWT without any fender, the bow suffers from severe structural damage. This is evident from the pronounced deformations not only in the watertight bulkheads, as shown in Fig. 18(a), but also in the surrounding hull structure, as depicted in Fig. 19(a). All internal reinforcements exhibit varying degrees of deformation, and slight warping is observed at the hull bottom. The maximum deformation of the 1st bulkhead occurs at its central region, while the 2nd bulkhead shows peak deformation near its connection to the deck. The recorded maximum deformation depths are 0.582 m and 0.431 m, respectively.

When the original metamaterial fender is applied, the overall bow response remains similar to the unprotected scenario as shown in Figs.

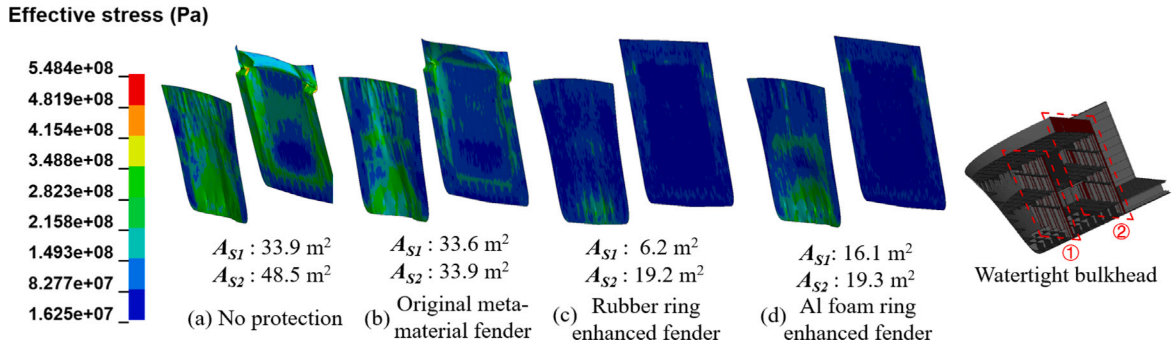


Fig. 18. Post-impact indentation and stress of watertight bulkheads in the bow.

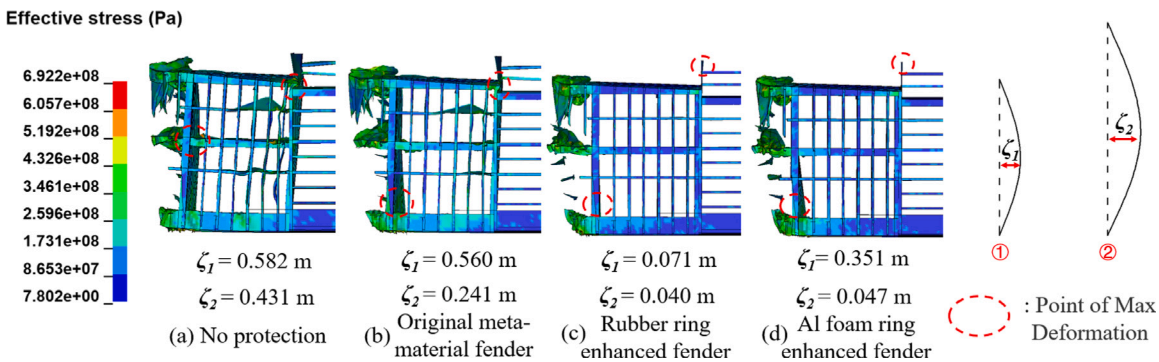


Fig. 19. Side view of post-impact indentation and stress of the internal structures of the impact bow.

18(b) and 19(b). However, notable improvements in localized deformation are observed. Specifically, the maximum deformation of the 1st bulkhead shifts to its bottom region, while the 2nd bulkhead sustains relatively less damage, exhibiting a reduced maximum crushing depth of 0.241 m.

In contrast, both ring-enhanced fenders provide markedly superior protection to the impacted vessel, as evidenced in Figs. 18(c)(d) and 19(c)(d). The most apparent improvement lies in the significant reduction of damaged areas and crushing depths in the watertight bulkheads. Both configurations effectively shield the 2nd bulkhead from severe deformation, limiting the maximum deformation depths to 0.040 m and 0.047 m, respectively. Notably, the rubber ring-enhanced fender offers superior protection for the 1st bulkhead as well. Furthermore, no appreciable stress concentrations or deformations are observed in the hull or internal reinforcements, indicating a successful mitigation of structural damage.

Although the watertight bulkheads did not exhibit visible broken holes in the simulations due to the simplified failure definition, it is important to note that, in actual collision accidents, structural failure holes are frequently associated with large deformations, potentially leading to water ingress. Such ingress can compromise the vessel's buoyancy and stability, increasing the risk of sinking. In contrast, both proposed ring-enhanced fender designs effectively limit the deformation of the 2nd bulkhead to a minimal level, thereby substantially reducing the likelihood of structural failure and enhancing the post-impact survivability of the vessel.

The evolution of stress distributions across different fender designs during the collision process is compared in Fig. 20. While all configurations exhibit similar developmental stages, they differ notably in deformation severity. Using the original metamaterial fender as a representative case, as illustrated in Fig. 20(a), several distinct phases of stress propagation and deformation are identified:

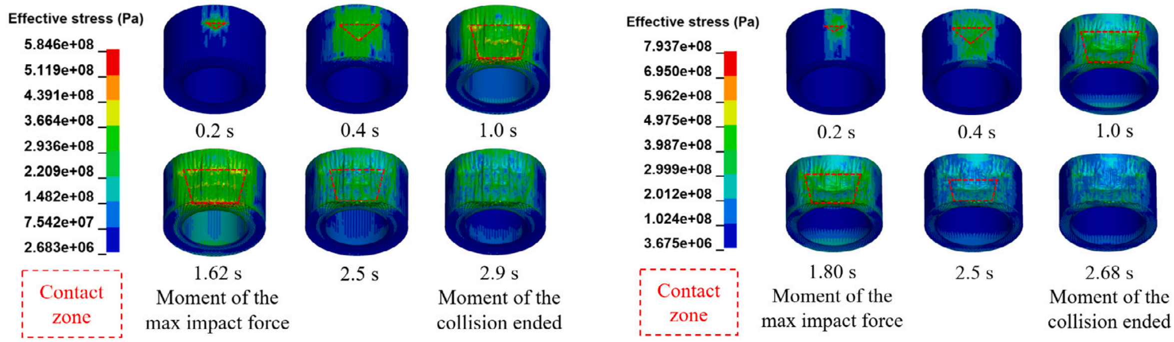
(1) 0–0.4 s: Bow initially contacts the fender, the stress propagates along both depth and radial directions, causing a local stress area which is larger than the bow-fender contact zone;

(2) 0.4–1.62 s (Moment of the max impact force): The main collision developing process, the impact force surges sharply until the maximum value, coupled with fluctuations due to the successive crushing behavior of the multiple metamaterial layers inside, as shown in Fig. 11. Simultaneously, due to the fast energy transformation from the vessel's kinetic energy to the structural internal energy and other dissipation mechanisms, as depicted in Figs. 12 and 13, the impact bow and fender both reach their maximum damage status; OWT itself also suffers from its elastic stress as shown in Fig. 15.

During this stage, bidirectional stress propagation within the metamaterial fender becomes increasingly pronounced. Besides the front local impact area, two distinct stress wave pathways emerge in the innermost layer, redirecting portions of the stress waves along the circular sides of the OWT, just like the cloaking functionalities [20]. This mechanism reduces the direct load transmitted to the OWT's front face and promotes a more distributed energy dissipation across the fender structure. As a result, local structural damage and deformation are effectively mitigated. Furthermore, owing to the integral nature of the fender and the substantial deformation experienced by its front external surface, the rear surface is also subjected to tensile stress, contributing to the overall energy absorption.

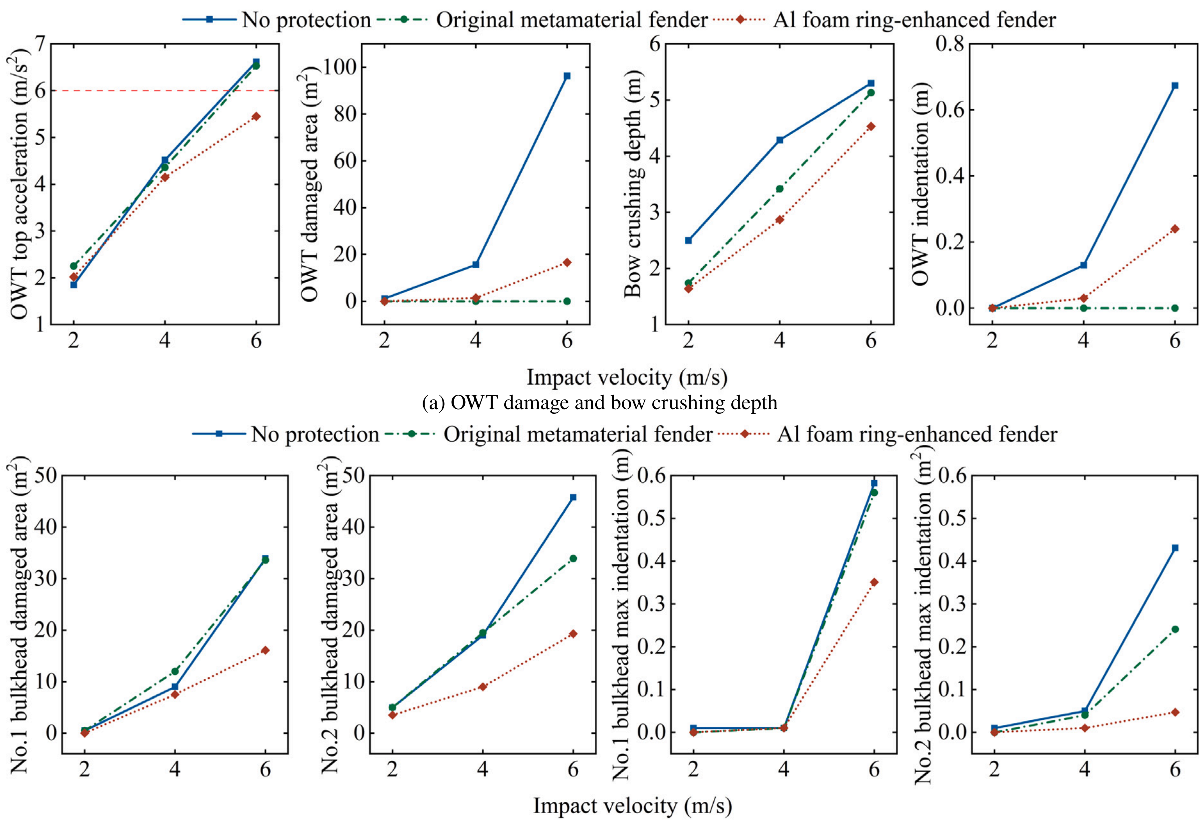
(3) 1.62 s - end of collision: In this final bow-fender separation phase, the vessel's residual kinetic energy is insufficient to support further penetration. Consequently, after a very brief quiescent period which manifested as a plateau of impact force in Fig. 11, the bow begins a slow and slight retraction. This is reflected by a minor drop of fender internal energy in Fig. 13, accompanied by a small increase in kinetic energy, attributed to the recovery of fender elastic deformation.

Both ring-enhanced fenders exhibit similar characteristics and stages with the original metamaterial fender. More details about the stress propagation process of the metamaterial components in ring-enhanced fenders are shown in Fig. 20(b) (The rubber ring-enhanced fender shows similarity to the Al ring-enhanced one, hence it is not illustrated for simplicity). Their stress development shares similar stages and trends with the original metamaterial fender in Fig. 20(a), but with



(a) Original metamaterial fender (Bottom-up view)
(b) Al foam ring enhanced fender (Bottom-up view)

Fig. 20. Stress and deformation development of the metamaterial part of various fender designs across the collision.



(a) OWT damage and bow crushing depth
(b) Watertight bulkhead damaged area and max deformation

Fig. 21. Post-impact response under various vessel impact velocities, 2~6 m/s.

improved energy dissipation and cloaking performance. Specifically, the metamaterial part of the rubber ring-enhanced fender and the Al foam ring-enhanced fender dissipate 54.2 MJ and 51.4 MJ of impact energy, respectively, exceeding the 45.5 MJ absorption capacity of the original metamaterial fender (see Fig. 13). This enhanced energy dissipation capability is evidenced by the larger deformation observed in both modified metamaterial parts compared to the original configuration. The presence of a 1 m hollow gap between the metamaterial part and the OWT at the fender's midsection facilitates this deformation, allowing for more progressive energy absorption.

The comprehensive evaluation of impact force, energy transformation, component damage patterns, and nacelle dynamic response confirms that both ring-enhanced fenders provide effective protection solutions. However, although the rubber ring-enhanced fender offers comparable protection to the vessel bow, it demonstrates inferior performance in safeguarding the OWT structure. Consequently, the Al foam ring-enhanced fender is selected as the optimal design, as it achieves synergistic performance by combining enhanced energy absorption, stress wave redirection (cloaking), and effective suppression of acceleration transmission. Furthermore, it delivers a well-balanced protective effect for both the OWT and the impacting vessel.

To improve computational efficiency, subsequent analyses will focus solely on the Al foam configuration.

5.2. Post-impact response under various impact velocities

Following comprehensive analysis under high vessel impact velocity, it remains valuable to investigate post-impact responses at lower impact speeds to assess the potential for practical reuse in engineering applications. Accordingly, numerical simulations are performed for three configurations, unprotected, original metamaterial fender, and Al foam ring-enhanced fender, under impact velocities of 2, 4, and 6 m/s.

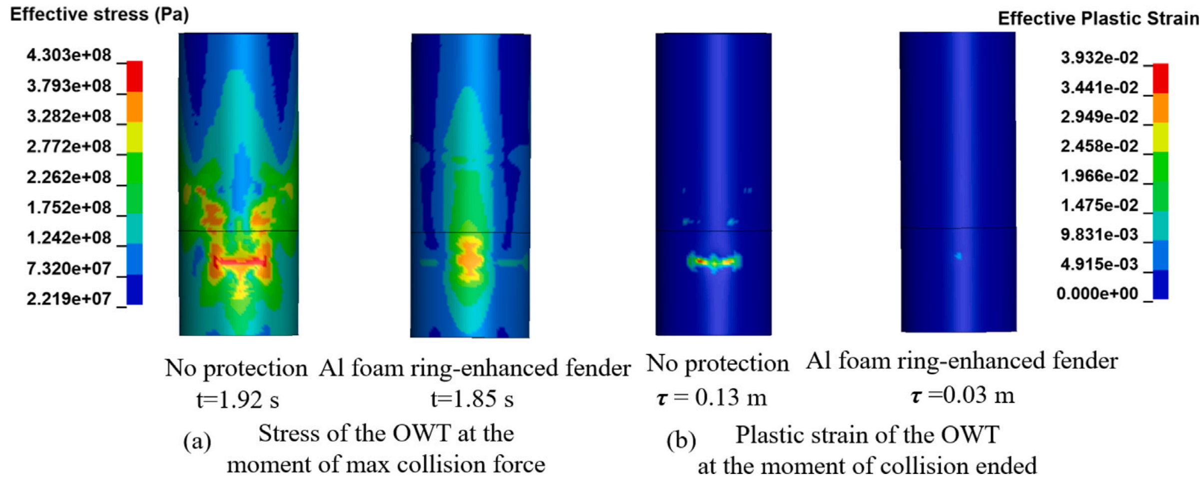


Fig. 22. Stress and plastic strain contour map of OWT under 4 m/s collision.

Key performance metrics are compared in Fig. 21. As illustrated in Fig. 21(a), a consistent increase in both the OWT top acceleration and the vessel bow crushing depth is observed with increasing impact velocities across all tested configurations. Among them, the Al foam ring-enhanced fender exhibits superior performance, reducing OWT nacelle acceleration by 10.2–16.5% and bow crushing depth by 5.7–11.7% compared to the original metamaterial fender acrossing the 2–6 m/s velocity range. Although the original metamaterial design provides a degree of structural protection, it fails to sufficiently suppress acceleration transmission; notably, at an impact velocity of 6 m/s, the peak acceleration approaches that of the unprotected scenario, exceeding the 6 m/s^2 safety threshold.

At 2 m/s impact velocity, no structural deformation is detected on the OWT across all configurations, shown in Fig. 21(a). When the impact velocity is elevated to 4 m/s, significant damage mitigation is achieved by both fender systems, with the OWT damaged area being reduced from 15.6 m^2 (unprotected baseline) to negligible levels.

For vessel bow deformation, crushing depth reductions of 16.2%~34.4% are achieved by the Al foam ring-enhanced fender across the impact velocity range. As presented in Fig. 21(b), superior performance is demonstrated by the enhanced fender system through significant reductions in both the watertight bulkhead damaged area and maximum deformation magnitude. Enhanced protective efficacy is observed at higher impact velocities.

A comparative analysis of the stress and plastic strain distributions is presented in Fig. 22 between unprotected and Al foam ring-enhanced fender cases at 4 m/s impact. The maximum effective stress in the OWT is reduced from 430 MPa (unprotected) to 372 MPa by the enhanced fender configuration, representing a 13.5% stress reduction. Correspondingly, the maximum effective plastic strain is decreased from 0.039 to 0.006, while structural indentation is reduced from 0.13 m to 0.03 m.

In addition, the deformation profiles of the Al foam ring-enhanced fender under varying impact velocities are presented in Fig. 23. As the impact velocity increases, both the depth and the lateral extent of the deformation region grow noticeably. However, under lower-velocity scenarios, the deformation remains localized primarily in the central region of the fender, indicating limited structural damage and the preservation of residual anti-collision capacity. This suggests that the design possesses favorable reusability and repairability characteristics, which are critical for practical engineering applications.

5.3. Post-impact response coupled with various wind conditions

The above analysis is conducted without considering wind loads, however, different combinations of wind loads and collision will cause various responses [5]. Three protection configurations (unprotected, original metamaterial fender, and Al foam ring-enhanced fender) are analyzed under 6 m/s vessel impact with concurrent wind loads (11 m/s and 25 m/s) from three directional orientations (0°, 90°, 180°).

Comparative post-impact responses are presented in Fig. 24, revealing several key findings: (i) The original metamaterial fender is demonstrated to completely eliminate OWT damage area and indentation across all wind load cases, though its acceleration mitigation capability is shown to be limited, particularly under wind-loaded conditions. (ii) Superior comprehensive performance is exhibited by the Al foam ring-enhanced fender, with simultaneous reductions being achieved in OWT structural damage, vessel structural damage and OWT top nacelle acceleration magnitudes.

Significant directional dependencies are revealed through comprehensive analysis: (i) At 0°, maximum acceleration peaks exceeding 6 m/s^2 are recorded across all configurations, attributed to constructive superposition of impact and wind energies, while OWT structural damage in both unprotected and Al foam ring-enhanced fender cases is minimized due to load path alignment; (ii) Under 180°, acceleration minima are observed but OWT structural damage except for original metamaterial fender cases is maximized, with local indentation being amplified by 96%–208% compared to wind-free cases - as opposing wind forces induce counteracting moments that increase deformation energy absorption; (iii) At 90°, negligible acceleration modulation is measured, but tower damage is exacerbated by 34.4–44.3% through coupled lateral loading mechanisms. The observed directional dependence of local indentation responses corroborates findings by [54], particularly regarding wind-induced moment effects on deformation mechanisms.

These response patterns persist across both wind speeds (11 m/s and 25 m/s), though enhanced rotor thrust effects at rated speed (11 m/s) are observed to intensify trend magnitudes of OWT structural damage and top acceleration. Notably, vessel bow crushing depth demonstrates remarkable insensitivity to both wind direction (<2.8% variation) and wind speed (<3.2% variation) across all protection configurations, indicating that bow deformation is primarily governed by impact kinematics rather than aerodynamic effects.

5.4. Post-impact response under varying metamaterial geometric configurations

This section discusses the influence of metamaterial geometric configurations on protective performance. Using a hierarchical design approach based on Eqs. (9) and (10), metamaterial configurations with varying numbers of elements per layer are systematically designed to

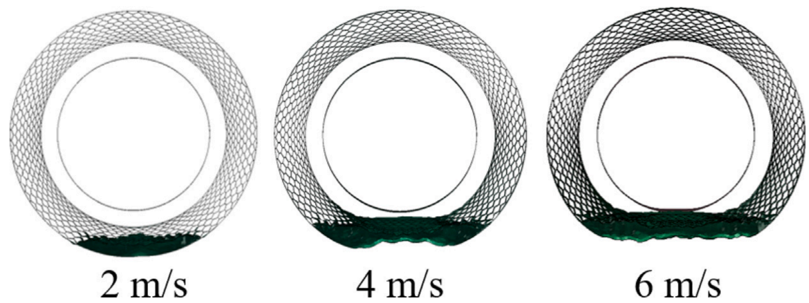


Fig. 23. Top view of deformations of the Al foam ring-enhanced fender under different impact velocities.

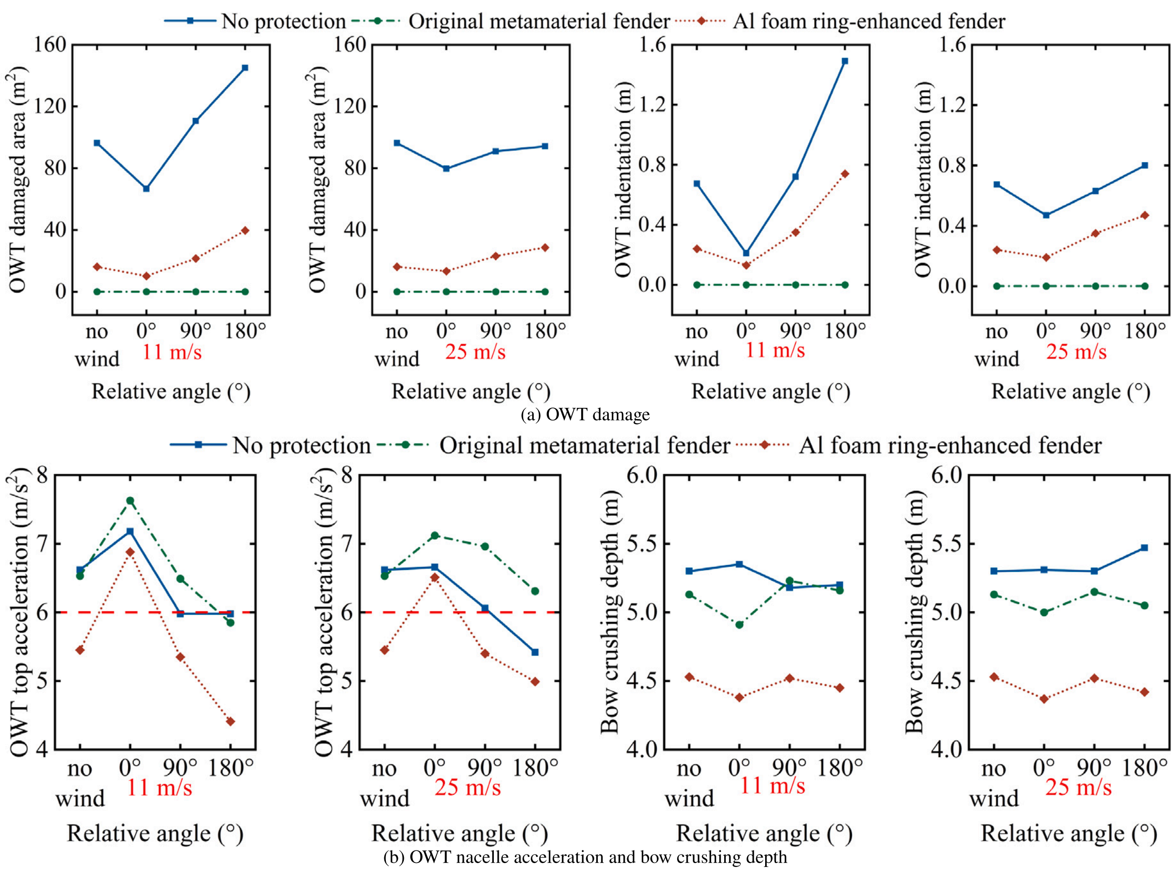


Fig. 24. Maximum post-impact response under 6 m/s head-on collisions coupled with wind loads.

examine the relationship between cell density and protection effectiveness. The investigation consists of two parts: (1) maintaining the original inner and outer diameters while modifying the number of cells in each layer, and (2) adjusting the inner and outer diameters while keeping the number of cells consistent with the initial design. Corresponding parameters and structural diagrams are provided in Fig. 25, wherein Fig. 25(c) represents the configuration selected as the research object in the preceding analysis.

A comparison of the protective performance across different geometric configurations in Fig. 25 subjected to the same ship impact is presented in Fig. 26. With a constant metamaterial layer thickness (δ_{meta} = 2 m) and varying cell density, OWT damage consistently decreases as cell density increases. Specifically, the damage area reduces by approximately 79.1% from 50 to 90 cells/layer, and the indentation depth is diminished by 82.4%. Conversely, the ship bow crushing depth increases by approximately 21.1% over the same range. This highlights a clear trade-off between OWT protection and ship safety: while higher metamaterial cell density enhances the structural integrity of OWT, it does so at the expense of increased vessel damage. Furthermore, the peak nacelle acceleration exhibits a non-monotonic trend: initially decreasing with cell density, reaching a minimum, and then rising again at 90 cells per layer, which approaches the design limit.

Consequently, achieving simultaneous minimization of damage to both the turbine structure and the vessel necessitates more than simply adjusting cell density. It is crucial to consider additional design parameters, such as plate thickness, fender height, and aluminum foam ring dimensions, alongside the metamaterial geometric properties. Future research will therefore focus on exploring the influence of these factors.

Extra comparison reveals that increasing δ_{meta} from 2 m to 3 m, while maintaining a constant cell density, leads to a 7.3% reduction in ship deformation and a 46.5% decrease in OWT damage. Moreover, increased cell density consistently yields a monotonic reduction in OWT permanent indentation and damage area. The configuration featuring a cell density of 72 cells/layer and δ_{meta} of 3 m demonstrates optimal energy management, effectively mitigating damage to both OWT and the ship. This optimal design limits permanent OWT indentation to

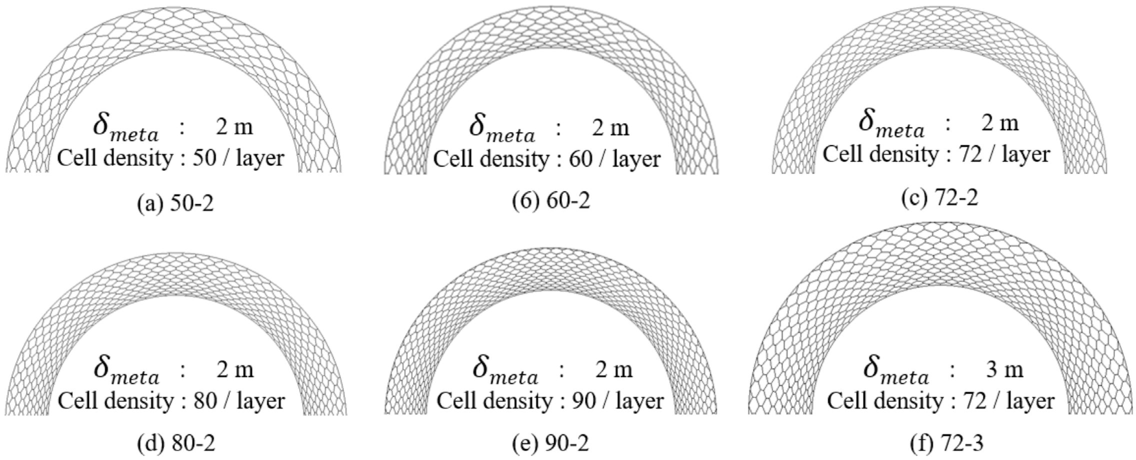


Fig. 25. Metamaterial structures under different design parameters.

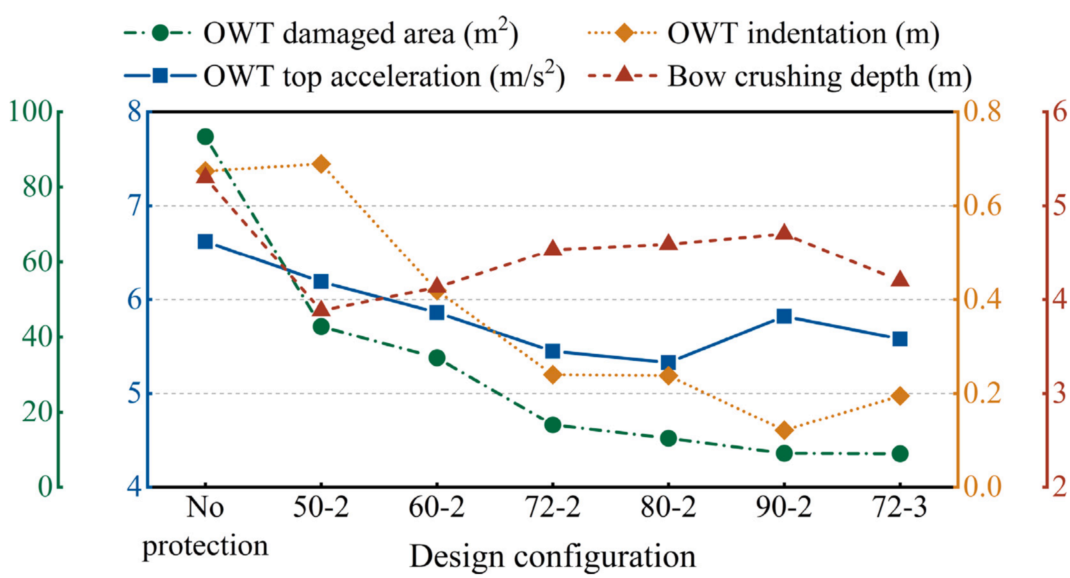


Fig. 26. Comparison of response metrics for various metamaterial configurations.

0.121 m, restrains the damage area to 8.962 m^2, and controls the bow crushing depth to 4.2 m.

However, practical implementation of such designs must balance navigational obstruction and economic feasibility. An increased fender diameter poses a greater risk of collisions with passing vessels, potentially leading to additional economic losses. Furthermore, increasing δ_{meta} from 2 m to 3 m raises the fender weight by approximately 35.3%, thereby elevating manufacturing costs and complicating installation. It is also worth noting that due to the radial layout of the metamaterial layers, increasing the diameter results in larger cell sizes in the outer regions, as shown in Fig. 25(c) and (f), which may partially counteract the protective benefits gained from diameter expansion.

6. Conclusions

This paper proposes a metamaterial-based fender concept for offshore wind turbines subjected to vessel impacts, with subsequent design modifications aimed at achieving a balanced protective performance for both the offshore wind turbine and the impacting vessel. Finite element simulations are employed to conduct detailed evaluations across multiple impact scenarios and design variables. The key conclusions are summarized as follows:

- The proposed metamaterial-based fender concept demonstrates effective protective performance and exhibits stress wave cloaking capabilities. The original metamaterial fender provides excellent protection against structural damage to the OWT under high-speed impacts (6 m/s); however, it offers limited suppression of nacelle accelerations and induces significant damage to the impacting vessel, potentially endangering crew safety. In contrast, the modified design, featuring the dual Al foam damping ring-enhanced meta fender, achieves a more balanced performance. It successfully limits nacelle accelerations within the predefined safety threshold and delivers simultaneous protection for both the OWT and the vessel.
- OWT–vessel collisions differ fundamentally from other waterway structure impact scenarios. The simulation duration must be sufficiently long to capture the system's response during the damped free vibration phase, and particular attention should be given to the safety of the impacting vessel.
- Under relatively low impact velocities (2–4 m/s), the fender maintains its protective functions for both the OWT and the vessel, while exhibiting favorable reusability and repairability—features that are crucial for practical engineering applications.

- A critical influence of wind direction on OWT response under vessel impacts is revealed. Peak accelerations (>6 m/s^2) are generated by co-directional winds (0°) through constructive superposition of impact and wind energies. Conversely, structural damage is maximized by counter-directional winds (180°) via wind-induced counteracting moments that enhance plastic deformation. Notably, consistency of these directional effects is observed across all protection configurations, though mitigation is provided by fender systems. The findings demonstrate that wind alignment must be considered in OWT collision risk assessments, particularly for extreme loading scenarios.

The limitations and directions for future work are acknowledged as follows: (i) Further investigation is required to elucidate the nonlinear relationships among OWT structural damage, nacelle acceleration, and fender stiffness, with the aim of refining the fender's performance. (ii) The present study omits wave-induced hydrodynamics, which are expected to influence monopile responses and alter ship motion behavior. These effects will be addressed in future research.

CRediT authorship contribution statement

Ming Song: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Jiyuan Chen:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Wenzhe Zhang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Javier Calderon-Sanchez:** Writing – review & editing, Supervision. **Zhiyu Jiang:** Writing – review & editing, Supervision.

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Declaration of competing interest

There are no conflicts of interest associated with the submission of this manuscript. All authors have reviewed and approved the manuscript for publication. On behalf of his co-authors, The author declare that the work presented is original research, has not been published previously, and is not under consideration for publication elsewhere, either in whole or in part. All listed authors have given their approval for the enclosed manuscript.

Data availability

Data will be made available on request.

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